# Su-Schrieffer-Heeger (SSH) Model

Dhruv Walia

July 2025

## 1 Introduction to the Su-Schrieffer-Heeger (SSH) Model

The SSH model is a one-dimensional tight-binding model that describes a chain of spinless fermions with alternating hopping strengths. It is one of the fundamental "toy models" for topological insulators in one dimension.

### 1.1 Hamiltonian of the SSH Model

In this model, we consider two sub-lattices, $A$ and $B$, with hopping only allowed between nearest neighbors. Two kinds of hopping are possible: hopping between $A$ and $B$ sites of the same unit cell, and hopping between neighboring unit cells. Let $v$ be the intra-cell hopping amplitude and $w$ be the inter-cell hopping amplitude.

The Hamiltonian can be expressed as:

$$H_{\text{SSH}} = v \sum_{n=1}^{N} |n, A\rangle \langle n, B| + w \sum_{n=1}^{N-1} |n+1, A\rangle \langle n, B| + \text{h.c.}$$

### 1.2 Hamiltonian in Terms of Internal and External Degrees of Freedom

There are two different types of degrees of freedom in this model: within the unit cell (internal) and between unit cells (external).

- **Internal Hopping:** This corresponds to hopping (from $A$ to $B$ and vice versa) within the same unit cell (same $n$).
- **External Hopping:** This corresponds to hopping (from $A$ to $B$ and vice versa) between adjacent unit cells (between $n$ and $n \pm 1$).

If $|n, \alpha\rangle \in \mathcal{H}$, we can decompose each state as $|n, \alpha\rangle = |n\rangle \otimes |\alpha\rangle$, where $|n\rangle \in \mathcal{H}_{\text{ext}}$, $|\alpha\rangle \in \mathcal{H}_{\text{int}}$, and $\mathcal{H} = \mathcal{H}_{\text{ext}} \otimes \mathcal{H}_{\text{int}}$.

Notice that $\mathcal{H}_{\text{int}}$ is spanned by the two states $|A\rangle$ and $|B\rangle$. Any $2 \times 2$ Hamiltonian involving only the internal degrees of freedom can be expanded in terms of the identity and Pauli matrices as $\sum_{i=0}^{3} d_i \sigma_i$, where $\sigma_0 = \mathbb{I}$ and $d_i \in \mathbb{R}$.

We can rewrite the SSH Hamiltonian as:

$$H_{\text{SSH}} = v \sum_{n=1}^{N} |n\rangle \langle n| \otimes \sigma_x + w \sum_{n=1}^{N-1} |n+1\rangle \langle n| \otimes \frac{\sigma_x + i\sigma_y}{2} + \text{h.c.}$$

or equivalently, using $\sigma_\pm = \frac{\sigma_x \pm i\sigma_y}{2}$:

$$H_{\text{SSH}} = v \sum_{n=1}^{N} |n\rangle \langle n| \otimes \sigma_x + w \sum_{n=1}^{N-1} |n+1\rangle \langle n| \otimes \sigma_+ + w \sum_{n=1}^{N-1} |n\rangle \langle n+1| \otimes \sigma_-$$

## 1.3 Translation Invariance and $H(k)$

In the large-$N$ limit, boundary effects (which break translation symmetry) can be neglected in the bulk. We can study the bulk Hamiltonian by imposing periodic boundary conditions, i.e., setting $|N+1\rangle = |1\rangle$:

$$\tilde{H}_{\text{SSH}} = \sum_{n=1}^{N} \left( v\ |n\rangle \langle n| \otimes \sigma_x + w\ |n+1\rangle \langle n| \otimes \sigma_+ + w\ |n\rangle \langle n+1| \otimes \sigma_- \right)$$

Due to the periodicity of the lattice, we can perform a Fourier transform to momentum space. This brings the Hamiltonian into a block-diagonal form, where each block corresponds to a specific crystal momentum $k$.

Let

$$|k\rangle = \frac{1}{\sqrt{N}} \sum_{n=1}^{N} e^{ink} |n\rangle, \quad \text{where } k = \frac{2\pi}{N} m,\ m \in \mathbb{Z},\ \text{and } -\frac{N}{2} < m \leq \frac{N}{2}.$$

The inverse transform is given by:

$$|n\rangle = \frac{1}{\sqrt{N}} \sum_{k'} e^{-ink'} |k'\rangle .$$

Substituting this into the bulk SSH Hamiltonian:

$$\begin{aligned}
\tilde{H}_{\text{SSH}} = v \sum_{n=1}^{N} & \left( \frac{1}{\sqrt{N}} \sum_{k'} e^{-ink'} |k'\rangle \right) \left( \frac{1}{\sqrt{N}} \sum_{k} e^{ink} \langle k| \right) \otimes \sigma_x \\
& + w \sum_{n=1}^{N} \left( \frac{1}{\sqrt{N}} \sum_{k'} e^{-i(n+1)k'} |k'\rangle \right) \left( \frac{1}{\sqrt{N}} \sum_{k} e^{ink} \langle k| \right) \otimes \sigma_+ \\
& + w \sum_{n=1}^{N} \left( \frac{1}{\sqrt{N}} \sum_{k'} e^{-ink'} |k'\rangle \right) \left( \frac{1}{\sqrt{N}} \sum_{k} e^{i(n+1)k} \langle k| \right) \otimes \sigma_- \\
= & \sum_{k} |k\rangle \langle k| \otimes \left( v\sigma_x + we^{-ik}\sigma_+ + we^{ik}\sigma_- \right).
\end{aligned}$$

Here we have used the orthogonality relation $\frac{1}{N}\sum_{n=1}^{N} e^{i(k-k')n} = \delta_{k,k'}$. The Bloch Hamiltonian in momentum space, $H(k) = \langle k| \tilde{H}_{\text{SSH}} |k\rangle$, is therefore given by:

$$H(k) = \begin{pmatrix} 0 & h(k) \\ h^*(k) & 0 \end{pmatrix},$$

where $h(k) = v + we^{-ik}$.

## 1.4 Energy Spectrum of $H(k)$

The energy eigenvalues $E_\pm(k)$ are found by diagonalizing $H(k)$:

$$E_\pm(k) = \pm|h(k)| = \pm\sqrt{v^2 + w^2 + 2vw\cos(k)}.$$

Clearly, $E_\pm(k) \neq 0$ for all $k$ whenever $v \neq w$. However, if $v = w$, the gap closes at the Brillouin zone boundary $k = \pm\pi$, where $E_\pm(\pm\pi) = 0$. Due to the chiral symmetry of the SSH model, a non-zero minimum $|E_\pm(k)|$ implies a symmetric band gap around $E = 0$, resulting in an insulating phase.

## 1.5 Lowest and Highest Energy States of $H(k)$

For a given $k \in (-\pi, \pi]$, we can write $h(k) = v + we^{-ik} \in \mathbb{C}$ in polar form as $h(k) = |h(k)|e^{i\arg(h(k))}$. The eigen-states corresponding to $E_\pm(k) = \pm|h(k)|$ are:

$$|\pm(k)\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm e^{-i\arg(h(k))} \end{pmatrix}.$$

Notice that $\arg(h(k))$ is only well-defined when $|h(k)| > 0$. Thus, this expression is ill-defined at the gap-closing point $v = w, k = \pm\pi$.

## 1.6 Winding Number

The topological invariant for the SSH model is the winding number $\nu$, defined as:

$$\begin{aligned} \nu &= \frac{1}{2\pi i} \oint_{\text{BZ}} \frac{1}{h}\, dh \\ &= \frac{1}{2\pi i} \int_{-\pi}^{\pi} \frac{1}{h(k)} \frac{dh(k)}{dk}\, dk. \end{aligned}$$

Evaluating this integral via Cauchy's residue theorem yields:

$$\nu = \begin{cases} 0 & \text{if } v > w \quad \text{(trivial phase)}, \\ 1 & \text{if } v < w \quad \text{(topological phase)}. \end{cases}$$

## 1.7 Understanding Symmetries in Quantum Systems

In quantum mechanics, a symmetry of a Hamiltonian $H$ is an operation represented by a unitary or anti-unitary operator $O$ such that:

$$OHO^{-1} = \pm H.$$

The sign in this equation distinguishes two types of symmetries:

- **Preserving symmetry:** $OHO^{-1} = H$, so $[H, O] = 0$.
- **Spectral-reversing symmetry:** $OHO^{-1} = -H$, so $\{H, O\} = 0$.

Four fundamental symmetries play a central role in topological band theory:

- **Chiral (Sublattice) Symmetry:** For a bipartite system built of two distinct sublattices, if there exists a unitary and Hermitian operator $\Gamma$ that anticommutes with the Hamiltonian, $\Gamma H \Gamma^{-1} = -H$ (or $\{H, \Gamma\} = 0$), the system possesses chiral symmetry.
- **Inversion (Parity) Symmetry:** Spatial reflection around the center of the system. If there exists a unitary and Hermitian operator $P$ such that $PHP^{-1} = H$, the system has inversion symmetry, mapping position $x \to -x$ and momentum $k \to -k$.
- **Particle-Hole (Charge Conjugation) Symmetry:** Represented by an anti-unitary operator $\mathcal{C} = U_C K$ (where $U_C$ is unitary and $K$ is complex conjugation) that transforms the Hamiltonian as $\mathcal{C}H\mathcal{C}^{-1} = -H$.
- **Time-Reversal Symmetry:** Represented by an anti-unitary operator $\mathcal{T} = U_T K$ that leaves the Hamiltonian invariant: $\mathcal{T}H\mathcal{T}^{-1} = H$. For spinless systems, $\mathcal{T}^2 = +1$, while for spin-1/2 systems, $\mathcal{T}^2 = -1$.

## 1.8 Symmetries of the SSH Model and Their Implications

While the periodic SSH chain has translation symmetry, translation symmetry is broken in a finite open chain and is not required to protect topological edge states. Instead, the SSH model belongs to symmetry class BDI, characterized by **chiral symmetry**, **time-reversal symmetry**, and **particle-hole symmetry**, along with spatial **inversion symmetry**.

### 1.8.1 Chiral (Sublattice) Symmetry

In the SSH model, hopping only connects sublattice $A$ to sublattice $B$, with no $A$–$A$ or $B$–$B$ terms. Consequently, the sublattice operator $\Gamma = \mathbb{I}_{\text{ext}} \otimes \sigma_z$ satisfies:

$$\{H, \Gamma\} = H\Gamma + \Gamma H = 0.$$

A direct consequence of chiral symmetry is that non-zero energy eigenvalues always appear in $\pm E$ pairs.
**Proof:** Suppose $|\psi\rangle$ is an eigenstate of $H$ with energy $E$:

$$H|\psi\rangle = E|\psi\rangle.$$

Applying $H$ to the state $\Gamma|\psi\rangle$ and using $H\Gamma = -\Gamma H$:

$$H(\Gamma|\psi\rangle) = -\Gamma H|\psi\rangle = -\Gamma(E|\psi\rangle) = -E(\Gamma|\psi\rangle).$$

Thus, if $|\psi\rangle$ is an eigenstate with energy $E$, $\Gamma|\psi\rangle$ is an eigenstate with energy $-E$.

### 1.8.2 Inversion (Parity) Symmetry

Reflecting the SSH chain about its center swaps the left and right halves of the chain as well as the $A$ and $B$ sub-lattices. The parity operator $P$ satisfies:

$$P^2 = \mathbb{I}, \quad P^\dagger = P = P^{-1}, \quad PHP^{-1} = H.$$

Because $P$ is both Hermitian and involutory, its eigenvalues are $\pm 1$, and simultaneous eigenstates of $H$ and $P$ have definite parity.

### 1.8.3 Time-Reversal Symmetry

For the spinless SSH model with real hopping amplitudes $v, w \in \mathbb{R}$, the time-reversal operator is simply complex conjugation, $\mathcal{T} = K$. Since $H_{\text{SSH}}$ is real, $\mathcal{T}H\mathcal{T}^{-1} = H^* = H$, with $\mathcal{T}^2 = +1$.

## 1.9 Edge States in the SSH Model

When we consider an open SSH chain, translation invariance is broken at the boundaries. Let us explicitly solve for localized boundary states.

### 1.9.1 Semi-Infinite SSH Model

Consider a semi-infinite SSH chain extending from $n = 1$ to $\infty$, with an open boundary at the left end ($n = 1$):

$$H_{\text{SSH}} = \sum_{n=1}^{\infty} \Big[ v\left(|n, A\rangle\langle n, B| + \text{h.c.}\right) + w\left(|n+1, A\rangle\langle n, B| + \text{h.c.}\right) \Big].$$

We expand a general eigenstate $|\psi\rangle$ in the site basis:

$$|\psi\rangle = \sum_{m=1}^{\infty} \left(\alpha_m |m, A\rangle + \beta_m |m, B\rangle\right).$$

Applying $H_{\text{SSH}}$ to $|\psi\rangle$ and using the orthonormality relation $\langle n, S|m, S'\rangle = \delta_{nm}\delta_{SS'}$ ($S, S' \in \{A, B\}$):

$$H_{\text{SSH}}|\psi\rangle = \sum_{n=1}^{\infty} v\left(\beta_n |n, A\rangle + \alpha_n |n, B\rangle\right) + \sum_{n=1}^{\infty} w\left(\beta_n |n+1, A\rangle + \alpha_{n+1} |n, B\rangle\right).$$

Re-indexing $n \to n-1$ in the $|n+1, A\rangle$ sum and enforcing the open boundary condition $\beta_0 = 0$:

$$H_{\text{SSH}} |\psi\rangle = \sum_{n=1}^{\infty} \left[ (v\beta_n + w\beta_{n-1}) |n, A\rangle + (v\alpha_n + w\alpha_{n+1}) |n, B\rangle \right].$$

Demanding $H_{\text{SSH}} |\psi\rangle = E |\psi\rangle$ yields the coupled recurrence relations:

$$\begin{aligned} (1) \quad & E\alpha_n = v\beta_n + w\beta_{n-1}, \\ (2) \quad & E\beta_n = v\alpha_n + w\alpha_{n+1}. \end{aligned}$$

Shifting Eq. (2) to $n-1$ gives $\alpha_n = \frac{E}{w}\beta_{n-1} - \frac{v}{w}\alpha_{n-1}$. Substituting this into Eq. (1) yields $\beta_n = -\frac{E}{w}\alpha_{n-1} + \frac{E^2 - w^2}{vw}\beta_{n-1}$. In matrix form:

$$\begin{bmatrix} \alpha_n \\ \beta_n \end{bmatrix} = M(E) \begin{bmatrix} \alpha_{n-1} \\ \beta_{n-1} \end{bmatrix}, \quad \text{with} \quad M(E) = \begin{bmatrix} -\dfrac{v}{w} & \dfrac{E}{w} \\ -\dfrac{E}{w} & \dfrac{E^2 - w^2}{vw} \end{bmatrix}.$$

Introducing the dimensionless variables $t = \frac{v}{w}$ and $x = \frac{E}{w}$, the transfer matrix is:

$$\mathcal{M} = \begin{bmatrix} -t & x \\ -x & \dfrac{x^2 - 1}{t} \end{bmatrix}, \quad \det(\mathcal{M}) = (-t)\left(\frac{x^2 - 1}{t}\right) + x^2 = 1.$$

Since $\mathcal{M} \in \text{SL}(2, \mathbb{R})$, its eigenvalues $\lambda_\pm$ satisfy $\lambda_+ \lambda_- = 1$ and are given by:

$$\lambda_\pm = \frac{\text{Tr}(\mathcal{M}) \pm \sqrt{\text{Tr}(\mathcal{M})^2 - 4}}{2}.$$

There are three regimes depending on $\text{Tr}(\mathcal{M})$:

- $\text{Tr}(\mathcal{M})^2 < 4$: $\lambda_\pm$ are complex conjugates of unit modulus ($|\lambda_\pm| = 1$), corresponding to extended bulk Bloch states.
- $\text{Tr}(\mathcal{M})^2 = 4$: $\lambda_+ = \lambda_- = \pm 1$, corresponding to the band edges. $\mathcal{M} = \pm\mathbb{I}$ occurs only when $x = 0$ and $t = \pm 1$ (the gap-closing transition point).
- $\text{Tr}(\mathcal{M})^2 > 4$: $\lambda_\pm$ are real and distinct, with $|\lambda_+| > 1 > |\lambda_-|$, corresponding to evanescent modes inside the energy gap.

Now let us look for the symmetry-protected zero-energy edge state by setting $E = 0$:

$$T := M(0) = \begin{bmatrix} -t & 0 \\ 0 & -\dfrac{1}{t} \end{bmatrix}.$$

The decoupled recurrence relations are $\alpha_n = -t\alpha_{n-1}$ and $\beta_n = -\frac{1}{t}\beta_{n-1}$. Using the boundary condition $\beta_0 = 0$, we immediately find $\beta_n = 0$ for all $n$, meaning the left edge state has support exclusively on sublattice $A$. For $\alpha_n$, we have $\alpha_n = (-t)^{n-1}\alpha_1$, giving:

$$|\psi\rangle = \alpha_1 \sum_{n=1}^{\infty} (-t)^{n-1} |n, A\rangle .$$

Normalizability ($\langle\psi|\psi\rangle = 1$) requires the geometric series $\sum_{n=1}^{\infty} |t|^{2(n-1)}$ to converge, which holds if and only if $|t| < 1$ (i.e., $|v| < |w|$). Normalizing the state yields:

$$|\psi\rangle = \sqrt{1-t^2} \sum_{n=1}^{\infty} (-t)^{n-1} |n, A\rangle .$$

The two-point correlation function between unit cells 1 and $m$ on sublattice $A$ is:

$$C(1, m) = \langle\psi|1, A\rangle \langle m, A|\psi\rangle = (1-t^2)(-t)^{m-1}.$$

Comparing $|C(1, m)| \propto e^{-|m-1|/\xi}$ defines the localization length:

$$\xi = -\frac{1}{\ln |t|},$$

which diverges as $t \to 1^-$ (the topological phase transition where the bulk gap closes).

## 1.10 Finite-Chain SSH Model

Consider a finite SSH chain of $N$ unit cells with open boundaries at both ends ($n = 1$ and $n = N$):

$$H_{\text{SSH}} = v \sum_{n=1}^{N} |n, A\rangle \langle n, B| + w \sum_{n=1}^{N-1} |n+1, A\rangle \langle n, B| + \text{h.c.}$$

In the topological regime ($|v| < |w|$) and for large $N$, the chain hosts two quasi-zero-energy edge states localized at the left and right boundaries (with an exponentially small hybridization splitting $\Delta E \propto (-t)^N$ that vanishes as $N \to \infty$). The state localized on the left edge is:

$$|L\rangle = \mathcal{N} \sum_{n=1}^{N} (-t)^{n-1} |n, A\rangle . \tag{1}$$

Since $N$ is finite, the second solution decaying from the right boundary ($n = N$) on the $B$ sublattice is also normalizable. We can obtain it using the spatial inversion (parity) operator $\mathcal{P}$, defined by:

$$\mathcal{P} |n, A\rangle = |N - n + 1, B\rangle ,$$
$$\mathcal{P} |n, B\rangle = |N - n + 1, A\rangle .$$

Applying $\mathcal{P}$ to $|L\rangle$ yields the right-localized edge state $|R\rangle = \mathcal{P}\,|L\rangle$:

$$|R\rangle = \mathcal{N}\sum_{n=1}^{N}(-t)^{n-1}\,|N-n+1,B\rangle = \mathcal{N}\sum_{m=1}^{N}(-t)^{N-m}\,|m,B\rangle\,, \tag{2}$$

where we changed variables to $m = N-n+1$. Both $|L\rangle$ and $|R\rangle$ share the same localization length $\xi = -1/\ln|t|$.

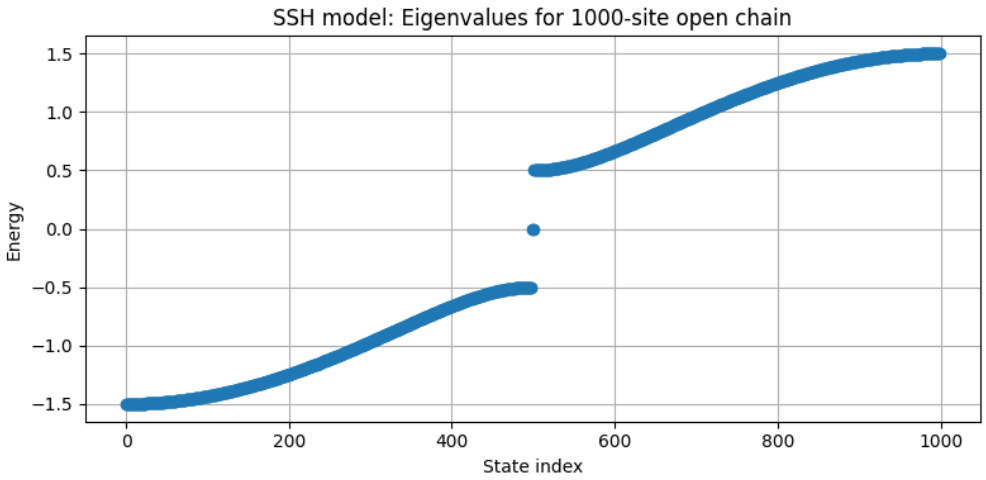
## 1.11 Numerical Calculation

To verify the existence and localization of edge states in the SSH model, we numerically diagonalize the tight-binding Hamiltonian for a finite chain of $N$ unit cells ($2N$ sites) with open boundary conditions. In the site basis $\{|1,A\rangle\,,|1,B\rangle\,,|2,A\rangle\,,\ldots,|N,B\rangle\}$, the $2N\times 2N$ Hamiltonian matrix takes the tridiagonal form:

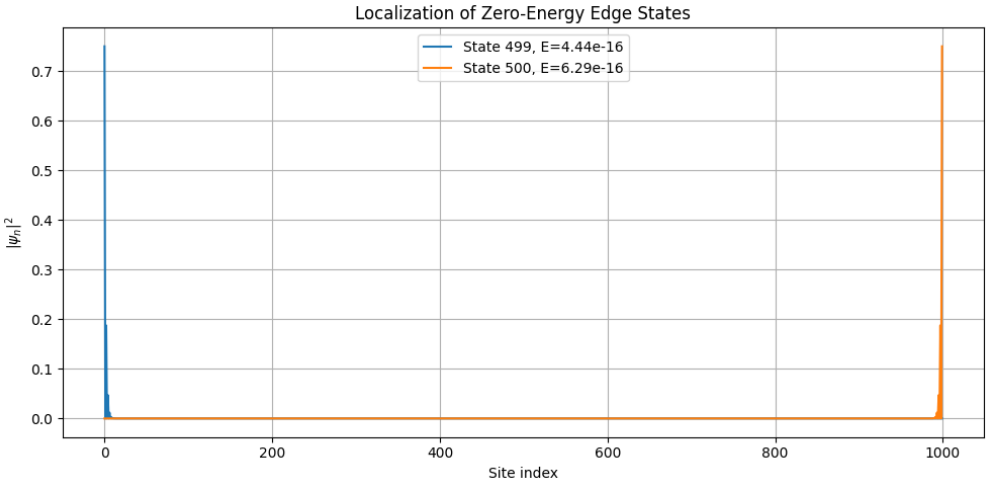
$$H_{\text{SSH}} = \begin{pmatrix} 0 & v & 0 & 0 & \cdots & 0 \\ v & 0 & w & 0 & \cdots & 0 \\ 0 & w & 0 & v & \cdots & 0 \\ 0 & 0 & v & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & v \\ 0 & 0 & 0 & 0 & v & 0 \end{pmatrix}.$$

Numerical diagonalization for $w > v$ reveals two mid-gap eigenvalues at $E \approx 0$, confirming the presence of topological edge states whose probability density $|\psi_n|^2$ is exponentially localized at the two ends of the chain.

These results confirm the bulk-boundary correspondence of the SSH model: when the bulk winding number is non-trivial ($w > v$, $|\nu| = 1$), the open chain hosts localized zero-energy edge modes. When $v > w$ ($|\nu| = 0$), these edge modes disappear and the spectrum is fully gapped.




(a) Eigenvalues of the SSH model with $v = 0.5$, $w = 1$, and $N = 1000$. Two zero-energy states are clearly visible.




(b) Probability density of the zero-energy states, showing localization at the edges.

Figure 1: Numerical results for the SSH model with open boundary conditions.